Oxford Mathematics Team Challenge 2026

Scenes Round Solutions

Act	Scene	A	B	C
I	1	2	27	19
	2	10	7	10
	3	2	2475	101
	4	288	6	3
II	1	4	9	12
	2	9	51	0
	3	10	108	20
	4	3304	12712	424

Act	I				II			
Scene	1	2	3	4	1	2	3	4
Points per question	3	4	5	6	3	4	6	9

Act I, Scene 1

Questions in this scene are worth 3 points

1A. What is the remainder when

$$2 + 4 + 6 + \cdots + 4052$$

is divided by 9?

SOLUTION

Let us factor out 2, and use the formula for the sum of natural numbers to get

$$2 + 4 + 6 + \cdots + 4052 = 2(1 + 2 + 3 + \cdots + 2026) = 2026 \cdot 2027$$

Now we can see that 2026, when divided by 9, has remainder 1, and 2027 has remainder 2. So our expression can be written as $(9m + 1)(9n + 2)$ for some positive integers m, n. Expanding out, we find the remainder $\boxed{2}$.

INVESTIGATION

Is it always true that the remainder of $m \cdot n$ when divided by k is the product of the remainders of m and n when divided by k? What if we take the remainder of the product of the remainders?

1B. The longest possible stick, of length ℓ m, is fitted into a sealed $1 \times 1 \times 1$ m^3 box. What is the value of ℓ^6?

SOLUTION

We observe that the largest stick inside a cube will be the line connecting opposite vertices.

To find this length, we note that it is the hypotenuse of a right triangle, with one side as an edge, and the other as the diagonal of a face. Using the Pythagorean theorem, we get that the length of the face diagonal is $\sqrt{2}$, and hence using the Pythagorean theorem again we get the length of our longest stick as
$\ell = \sqrt{(\sqrt{2})^2 + 1^2} = \sqrt{3}$.

Alternatively, placing the cube in the first octant of $3d$ space, with a vertex at the origin, the opposite vertex will be at $(1, 1, 1)$. Thus the distance is
$\ell = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$

Ergo, $\ell^6 = 3^3 = \boxed{27}$

1C. A drawer has ten pairs of socks, all of different colours. Two socks are randomly picked from the drawer. The probability that the socks are of the same colour is p. What is $1/p$?

SOLUTION

After we pick the first sock, there is precisely one other sock that is the same colour out of the 19 that remain. Hence there is a $\frac{1}{19}$ probability that you take the matching pair.

Alternatively, there are ${}^{20}C_2 = 190$ ways to pick 2 socks and there are 10 pairs that work, so the probability is $\frac{10}{190} = \frac{1}{19}$. Hence $\frac{1}{p} = \boxed{19}$

Act I, Scene 2

Questions in this scene are worth 4 points

2A. When the expression

$$\left(x + \frac{1}{x}\right)\left(x^2 + \frac{2}{x^2}\right)\left(x^3 + \frac{3}{x^3}\right)\left(x^4 + \frac{4}{x^4}\right)$$

is expanded and like terms are collected, one term is independent of x. What is its value?

SOLUTION

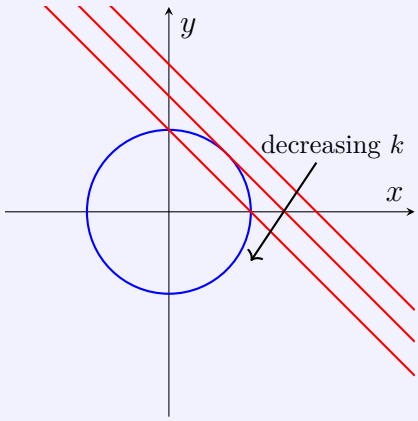
First, we factor out $1/x^{10}$ to get $\frac{1}{x^{10}}(x^2+1)(x^4+2)(x^6+3)(x^8+4)$;

The constant term needs the powers to sum up to x^0, so the factors in the brackets need to multiply up to a multiple of x^{10}. This is possible only by multipling x^2 and x^8 or x^4 and x^6, giving us $2 \times 3 + 1 \times 4 = \boxed{10}$.

2B. Suppose x and y are real numbers such that $x^2 + y^2 = 25$. The maximum possible value of $x + y$ is $\frac{a}{\sqrt{b}}$ when fully simplified. What is $a + b$?

SOLUTION

We consider the graph of $x^2 + y^2 = 25$ and $x + y = k$ for varying values of k. For large values of k, these graphs don't intersect. We can see that the first value of k for which they do intersect is the one where the line is tangent to the circle.

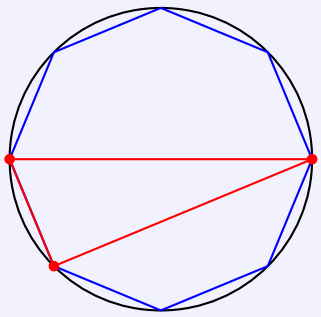


This tangent hits the circle at $\left(\frac{5}{\sqrt{2}}, \frac{5}{\sqrt{2}}\right)$ (by symmetry in x and y), thus $k = 5\sqrt{2}$, and $a + b = 5 + 2 = \boxed{7}$

2C. Octo the Octopus randomly chooses three distinct vertices from a regular octagon. When fully simplified, the probability that the triangle formed from these vertices is a right angle is p/q. What does $p + q$ equal?

SOLUTION

Inscribing our octagon in a circle, we see that our triangle is right if and only if we choose two vertices along a diameter.



There are 4 diameters to choose and 6 other vertices, so 24 right triangles. There are $\frac{8 \cdot 7 \cdot 6}{3!} = 56$ ways to choose a triangle, so our probability is $\frac{24}{56} = \frac{3}{7}$. Our answer is therefore $3 + 7 = \boxed{10}$.

INVESTIGATION

What would be the probability for a regular n-sided polygon?
[Hint: consider odd and even n separately]

Act I, Scene 3

Questions in this scene are worth 5 points

3A. The two graphs

$$y = 8 - x^3, \qquad y = 2x^3 - 9x^2 + 20x - 20$$

intersect at exactly one point, (a, b). What does a equal?

SOLUTION

We factor the polynomials as $(x-2)(x^2+2x+4)$, $(x-2)(2x^2-5x+10x)$, respectively. Hence both curves intersect at $x = 2$ (at $(2, 0)$). Hence $a = \boxed{2}$

3B. Carl writes down every positive integer less than 100 whose digits sum to an even number. What is the sum of Carl's numbers?

SOLUTION

Any such number has either two odd or two even digits (including a leading 0 as an even digit). Each number from 0-9 is used precisely 5 times each in the tens digit and ones digit. Decomposing the answer into ones and tens, the answer is $5(0 + 10 + 20 + ... + 90) + 5(0 + 1 + 2 + ... + 9) = 55 \cdot 45 = \boxed{2475}$

3C. Suppose α is a number satisfying $\alpha^2 + 2\alpha + 4 = 0$. It follows that $\alpha^{99} = x^y$, where x is a prime number and y is an integer. What is $x + y$? *[Hint: you do not need to calculate α.]*

SOLUTION

We note that $(\alpha - 2)(\alpha^2 + 2 + 4) = 0$, i.e. $\alpha^3 - 8 = 0$. That is, $\alpha^3 = 2^3$, so $\alpha^{99} = 2^{99}$. Thus $x + y = \boxed{101}$.

Act I, Scene 4

Questions in this scene are worth 6 points

4A. Mrs Fifield fills in a 4×4 grid with four 1s, four 2s, four 3s and four 4s. She does it in such a way that each column, each row and the four sub-squares in the diagram each contain the numbers $1, 2, 3, 4$. For example:

2	4	3	1
3	1	2	4
1	3	4	2
4	2	1	3

How many different ways could Mrs Fifield have filled in the grid?

SOLUTION

Note that if we fill in the grid, and then permute the numbers $1, 2, 3, 4$, we get another solution. So, let us fix the numbers $1, 2, 3, 4$ in the top row, find the number of grids in this scenario, and can then multiply by 4! to get the total number of grids.

Now, there are 4 possibilities in the first column. We fill first $1, 3, 2, 4$ and find that it gives us 3 unique solutions, drawn below:

1	2	3	4
3	4	1	2
2	1	4	3
4	3	2	1

1	2	3	4
3	4	2	1
2	1	4	3
4	3	1	2

1	2	3	4
3	4	1	2
2	3	4	1
4	1	2	3

Flipping the 2 and 4 gives us a 3 different unique solutions. Similarly 1, 4, 2, 3 and 1, 4, 3, 2 in the first column give us six more solutions. So in total we have 12 solutions in this scenario, multiplying by 4! we get $\boxed{288}$.

4B. Anna has the functions

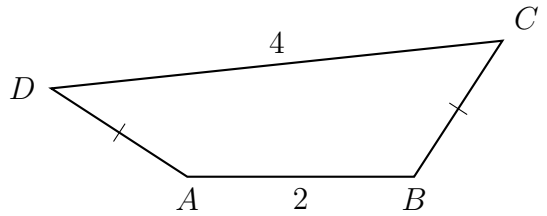
$$r(x) = 1/x \qquad \text{and} \qquad s(x) = 1 - x.$$

She starts with some number n, and repeatedly applies these two functions in order to obtain as many distinct numbers as she can. What is the largest number of distinct numbers she can make, including n?

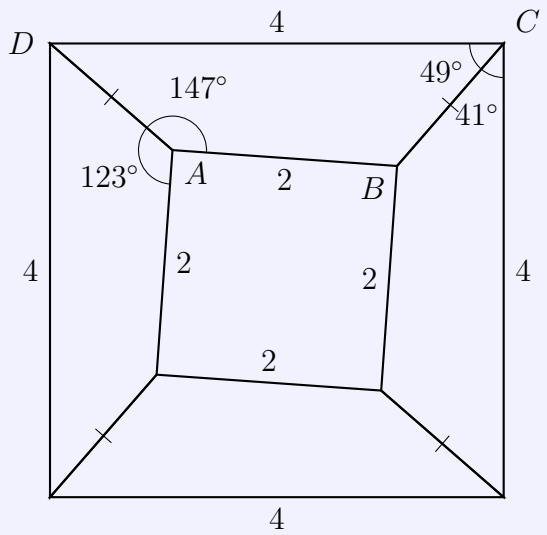
SOLUTION

First note that $r(r(n)) = n = s(s(n))$. Hence, to get more numbers besides n, $r(n)$, and $s(n)$, we must alternate applying r and s to get different numbers. However, $r(s(r(n))) = s(r(s(n)))$ so there are at most 6 numbers, being $\{n, r(n), s(r(n)), s(n), r(s(n)), r(s(r(n)))\}$. Taking $n = 3$, for example, shows that we can obtain $\{3, -2, 1/3, -1/2, 3/2, 2/3\}$, so $\boxed{6}$ is the attainable maximum.

4C. Consider a quadrilateral $ABCD$ with the lengths $AB = 2$, $CD = 4$, $BC = AD$, and $\angle DAB = 147°$, $\angle ABC = 123°$, and $\angle BCD = 49°$. Find the area of $ABCD$.



SOLUTION



As seen in the diagram, we take 4 copies of the quadrilateral, rotate them, glue together to get a square of side length 4 with a square hole of side length 2. (Note that the angles at each edge of the big square add up to $49° + 41° = 90°$, and the outer angles of the inner square add up to $123° + 147° = 270°$, so they are infact squares). Hence the area for 4 of the quadrilaterals is $4^2 - 2^2 = 16 - 4 = 12$, so one of them has area $\boxed{3}$.

Act II, Scene 1

Questions in this scene are worth 3 points

1A. 'Gnatty Nety' thinks of a positive integer n such that n is not prime and n is not a factor of $(n-1)!$.

What is the largest number that Nety could've thought of?

SOLUTION

$n = \boxed{4}$ clearly works, and it is infact the answer.

Every non-prime number greater than 4 can be written as $n = a \cdot b$ with either a or b strictly greater than 2 but strictly less than n. If a and b are different, then they appear in the product for $(n-1)!$. If $a = b$, then since $b > 2$, $2a$ appears in the product for $(n-1)!$, and hence n divides $(n-1)!$.

1B. For $0^\circ \leq x \leq 360^\circ$, how many solutions does the equation

$$\sin(2x + 90^\circ) + \cos(2x - 90^\circ) = 1$$

have?

SOLUTION

Square both sides to get

$$\sin^2(2x + 90^\circ) + 2\sin(2x + 90^\circ)\cos(2x - 90^\circ) + \cos^2(2x - 90^\circ) = 1$$

Since $\cos(2x - 90^\circ) = -\cos(2x + 90^\circ)$, so $\cos^2(2x - 90^\circ) = \cos^2(2x + 90^\circ)$, so the equation yields

$$2\sin(2x + 90^\circ)\cos(2x - 90^\circ) = 0$$

Only one of the two can be 0 at a specific angle. If $\sin(2x + 90^\circ) = 0$ then there are four solutions; if $\cos(2x - 90^\circ) = 0$ then there are five solutions. However only 5 of these solutions give us 1 when plugged into the original equation, the rest give -1. Thus there are $\boxed{5}$ solutions in total.

1C. Quincy chooses a, b, c to be whole numbers in the set $\{1, 2, 3, 4\}$, potentially choosing the same values. The quadratic

$$ax^2 + bx + c$$

has at least one real root. How many choices of (a, b, c) could Quincy have chosen?

SOLUTION

A quadratic has at least one real root if its discriminant is non-negative. We condition on b, and count case-by-case.

- $b = 1$ has no such (a, c)
- $b = 2$ has $(a, c) = (1, 1)$
- $b = 3$ has all of the above and $(a, c) = (1, 2)$, $(2, 1)$
- $b = 4$ has all of the above and $(a, c) = (1, 3)$, $(3, 1)$, $(1, 4)$, $(4, 1)$, $(2, 2)$

Thus in total we have $1 + (1 + 2) + (1 + 2 + 5) = \boxed{12}$ solutions.

Act II, Scene 2

Questions in this scene are worth 4 points

2A. A fake scoop of ice cream is used for a display and is modelled as follows. The cone has a base radius of R and height $3R$; placed on top of it is a hemisphere of radius R, glued onto the base of the cone. The cone is then cut horizontally to remove a cone with a base radius kR, so that the volume of the remaining model is twice that of the volume of the hemisphere. What is k^{-6}?

SOLUTION

The volume of the cone is $\frac{1}{3}\pi R^2(3R) = \pi R^3$, volume of the hemisphere is $\frac{2}{3}\pi R^3$.

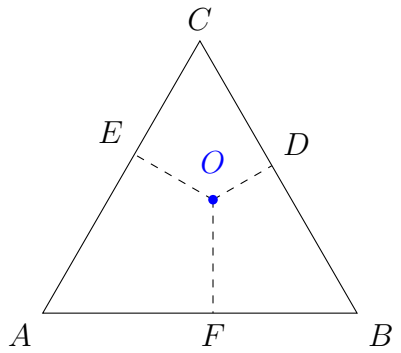
Cutting the cone horizontally, the removed cone of base radius kR will have height $3kR$. One can see this by taking half of the central cross section of the cone giving a right triangle with one side length kR. The tangent of the angle at the vertex of the cone is the same as in the original cone, so the ratio of the two sides is the same. So volume removed is $\pi k^3 R^3$.

The remaining part of the original cone must have volume $\frac{2}{3}\pi R^3$, so we have

$$\pi R^3 - \pi k^3 R^3 = \frac{2}{3}\pi R^3 \Rightarrow k = 3^{-\frac{1}{3}}.$$

Thus $k^{-6} = \boxed{9}$

2B. Let ABC be an equilateral triangle with a point O in its interior. The point O is put inside the triangle and three lines are drawn to meet each side of the triangle perpendicularly, at the points D, E and F. Given $OD = 3$, $OE = 4$ and $OF = 5$, it follows that the area of ABC is $a\sqrt{b}$ when fully simplified. What is $a + b$?



SOLUTION

Letting the side length equal s. We draw the lines AO, BO, CO. The area of $\triangle AOB$ is $\frac{1}{2} \cdot s \cdot 5$. We similarly find the areas of $\triangle AOC$ and $\triangle BOC$.

Adding these up, and using the formula for the area of an equilateral triangle, we get

$$\frac{1}{2}(3 + 4 + 5)s = \frac{\sqrt{3}}{4}s^2$$

Solving this for $s \neq 0$, we get $s = 8\sqrt{3}$, so our area is $48\sqrt{3}$, giving us an answer of $48 + 3 = \boxed{51}$.

2C. How many integer solutions (x, y) are there to the equation

$$x^3 - 117y^3 = 5?$$

[Hint: what are the remainders of x^3 when divided by 9?]

SOLUTION

Checking 0, 1, and so on, one sees that the only remainders of x^3 when divided by 9 are 0, 1, and 8. Finding the remainders of both sides of the equation when divided by 9, since 117 is divisible by 9, we get that the remainder when x^3 is divided by 9 is 5. Since this is not possible, there are no solutions to the equation. Thus our answer is $\boxed{0}$

Act II, Scene 3

Questions in this scene are worth 6 points

3A. Robin the Robot rolls around a rectangular $m \times n$ grid of rooms. He visits each room exactly once and ends up in a room next to the room he started in.

Given that there are between 10 and 20 rooms, inclusive, and $m \leq n$, how many different ordered pairs (m, n) could Robin be rolling around?

SOLUTION

Robin has to roll around an even number of rooms - if we chequer-pattern them black and white, Robin alternates between black and white, and to loop the last room he's in must be of the opposite colour of the one he started in.

In order to be able to loop, $m, n \geq 2$. Other than the fact that $10 \leq mn \leq 20$, we have no restrictions. The possible room sizes are (2, 5), (2, 6), (3, 4), (2, 7), (2, 8), (4, 4), (2, 9), (3, 6), (2, 10), (4, 5). So there are $\boxed{10}$ different room sizes.

3B. Danny has a dataset $\{5, 5, 7, 8, 12, 12\}$. He wants to add two (not necessarily distinct) positive integers x, y to it such that there is a unique mode which is also equal to the mean. What are the sum of the values of $x + y$ for all possible ordered pairs (x, y) that Danny can add to his dataset?

SOLUTION

If $x \neq y$, then one of x, y must be 5 or 12. Testing these two cases, we see that if $x = 5$, y will be negative, so this case gives us the answers $(12, 35), (35, 12)$.

If $x = y$, then to preserve a unique mode, $x = y$ must be in $5, 7, 8, 12$. Testing these cases we see that none of them work.

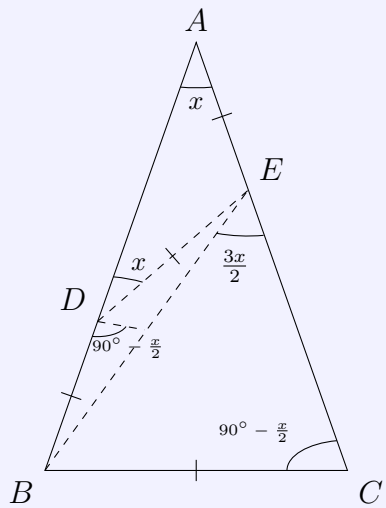
Hence, the answer is $12 + 35 + 35 + 12 = \boxed{94}$

3C. Consider an isosceles triangle ABC with $AB = AC$, and let D, E be points on AB and AC respectively such that

$$AE = DE = DB = BC.$$

Find the angle $\angle BAC$ in degrees. *[Hint: draw the line EB.]*

SOLUTION



Let $AE = 1, x = \angle BAC$. Angle chasing as above, we find that $BE = 2 \cdot \sin\left(90^\circ - \frac{x}{2}\right) = 2\cos\left(\frac{x}{2}\right)$.

We also find $\angle DEC = 2x, \angle DEB = \frac{x}{2}$ so $\angle BEC = \frac{3x}{2}$. And $\angle BCA = 90^\circ - \frac{x}{2}$

Using the sine law on $\triangle BEC$,

$$\frac{\sin\left(\frac{3x}{2}\right)}{1} = \frac{\sin\left(90^\circ - \frac{x}{2}\right)}{2\cos\left(\frac{x}{2}\right)} \Rightarrow \sin\left(\frac{3x}{2}\right) = \frac{1}{2}$$

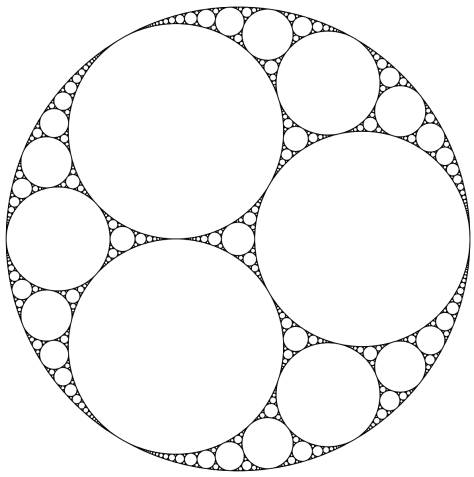
so $\frac{3x}{2} = 30^\circ$ or $150^\circ \Rightarrow x = 20^\circ$ or 100°. However note that 100° is impossible as $|AC| > |BC|$, so $x = \boxed{20^\circ}$.

Act II, Scene 4: Estimathon

Questions in this scene are worth up to 9 points: where A is the actual answer and E is your estimate for a question, your score is $9 \times \frac{\min(A,E)}{\max(A,E)}$ points, rounded to the nearest integer

4A. The *Apollonian Gasket* is a fractal, built by considering recursively the inner tangent circle to three kissing circles (mutually touching circles).

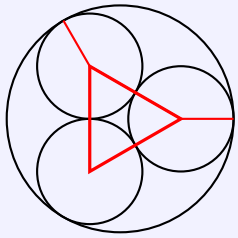
Given that the outermost circle has radius $r = 1$, how many circles in the Gasket will have a radius $r \geq 0.001$?



SOLUTION

Call the outermost circle C_0. First, we should find the radii of the jointly second-largest circles (call these circles C_a, C_b, C_c:

Letting these circles have radius r, we can calculate that the distance from one of their centres (i.e. one of the vertices of the triangle) to the centre of C_0 is $r/\cos 30° = 2r/\sqrt{3}$. It follows that $r(1 + 2/\sqrt{3}) = 1$ so $r = 2\sqrt{3} - 3$ (which is roughly 0.5 as $r = \sqrt{12} - \sqrt{9}$, and 12 is roughly halfway between 9 and 16).



We state here Descartes' Circle Theorem for mutually tangent circles. We define the curvature of a circle $k = \pm 1/r$. The $\pm$ in k is taken to be positive or negative, depending on whether we want an externally tangent or internally tangent circle respectively.

The circle of curvature k_4 which is mutually tangent to the circles k_1, k_2, k_3 is related by:

$$k_4 = k_1 + k_2 + k_3 + 2\sqrt{k_1k_2 + k_2k_3 + k_1k_3}$$

In our case, initially we have 4 circles with $k \approx -1, 2, 2, 2$ (We take -1, as the big circle is internally tangent). We can use this formula to calculate iteratively the curvature of all the circles formed in the gasket, and stop when we get to $k = 1000$. However, we don't have so much time, so lets approximate!

Let us focus in on one of the 3 outer areas. Here we calculate the curvature of the circle made on each pass, and list them below. Here we have only calculated those with $k \leq 30$.

1: 3; 2: 6, 6, 15; 3: 11, 11, 14, 14, 23, 23; 4: 18, 18, 26, 26; 5: 27, 27; 6: 30, 30

We also look at the central area and find the (approximate) first few circles.

1: 13; 2: 32, 32, 32

Let $N(k)$ denote the number of circles of curvature $\leq k$. We assume an exponential growth $N(k) = Ak^x$, where A and x are to be determined.

By the above data we find $N(30) = 3 \cdot 19 + 1 = 58$, $N(15) = 3 \cdot 9 + 1 = 28$.

So

$$\frac{58}{28} = \frac{N(30)}{N(15)} = \left(\frac{30}{15}\right)^x$$

$$\Rightarrow x = \frac{\log(58/28)}{\log(2)} = \frac{\log(2) + \log(29/28)}{\log(2)} = 1 + \frac{\log(1 + 1/28)}{\log(2)} \approx 1 + \frac{1/28}{0.3} \approx 1.12$$

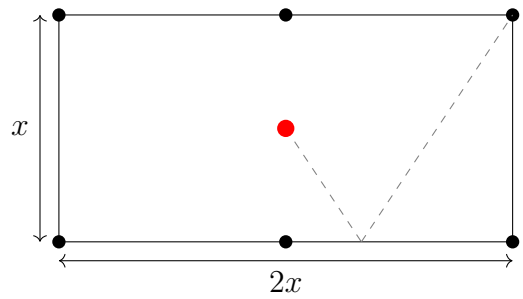
Now

$$\frac{N(1000)}{N(30)} = \left(\frac{1000}{30}\right)^{1.12} \approx 33^{1.12} \approx 33 \cdot 32^{0.12} = 33 \cdot 2^{0.6}$$

Since $2^{0.5} \approx 1.414$ we will take $33 \cdot 2^{0.6} \approx 50$. Thus $N(1000) \approx 58 \cdot 50 = 2900$.

Note that our initial estimates are expected to be erroneous (and low) compared to the long term growth. The actual answer is $\boxed{3304}$.

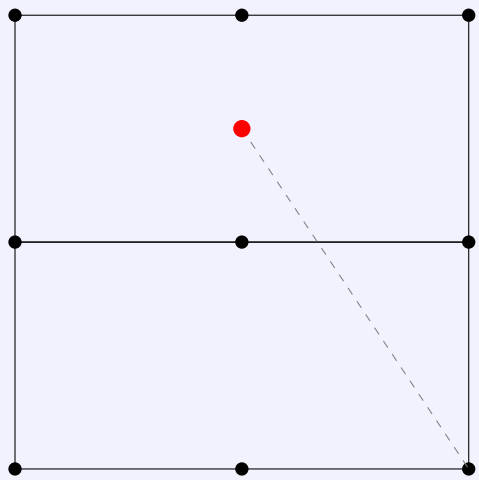
4B. Billy picks an angle to strike a red ball from on the billiards table shown below. Each bounce reflects the ball's trajectory against the line of the table, and after exactly 2026 bounces the ball goes into one of the six holes. How many angles could Billy have picked? *[Below is an example of the ball going into a hole after one bounce.]*



SOLUTION

We first demonstrate a (miraculous!) exact solution to this problem. At the end, we try to approximate this.

One way to make the calculation easier is to reflect the table itself along its sides repeatedly, to get a lattice of tables. Then the ball going into a whole after n bounces is the same as the ball on our infinite grid passing through n lines before landing on one of the pockets.



For example, the demonstration above shows the equivalent trajectory for the diagram in the question on our grid like structure. We can see that reflecting the trajectory over one of the sides, gives us a straight-line path on our infinite grid.

We'll note that also if we just solve the number of trajectories that initially start with the ball heading towards the top-right pocket (ie with trajectory in the first quadrant) accounts for exactly a quarter of the total number of trajectories that gets sunk in 2026 bounces. **So we will exclusively work in the top right quadrant and multiply our final answer by 4**.

Now affine transformations preserve reflections, ie if we label the holes of our original table on our plane as $(1,1)$, $(1,0)$, $(1,-1)$, $(-1,1)$, $(-1,0)$, $(-1,-1)$, and expanding by reflecting the table over itself repeatedly, the holes with coordinates (s,t) that represent sinking the ball which starts on $(0,0)$ in 2026 bounces are given by:

$$\frac{s-1}{2}+\frac{t-1}{2}=2026 \text{ (for corner pockets)}$$

and

$$\frac{s-1}{2}+\frac{t}{2}=2026 \text{ (for side pockets).}$$

Simplifying these equations, we get:

$$s+t=4054 \text{ (for corner pockets)}$$
$$s+t=4053 \text{ (for side pockets).}$$

But how would we know whether the ball is actually still in play before it gets sunk into a pocket after 2026 bounces (it could've been pocketed before)? Well, noticing that our trajectories are given by straight lines, and our holes, by construction, must have coordinates (a,b) where both a and b are integers. Even more so, they must be in the form of $(odd, even)$ and (odd, odd) and also all points with these coordinates are necessarily holes.

So we find that actually all we need are our final hole coordinates (s,t) to be co-prime (they share no common factors) and are also either of the form $(odd, even)$ or (odd, odd). This is because we can model the trajectory by the line $y=\left(\frac{t}{s}\right)x$, so if $\frac{t}{s}=\frac{p}{q}$ where $|p|<|t|$ (ie we can reduce the fraction), then (q,p) is a lattice point that this line passes through. But also notice that q must have the same parity as s and p must have the same parity as t, so if (s,t) is a hole (q,p) must be a hole as well.

Now for solving for (s,t) where $s+t=4054$, s, t co-prime, and (s,t) are in the form $(odd, even)$ or (odd, odd), we must first have:

$$\begin{aligned}1 &= gcd(s,t)\\ &= gcd(s,4054-s)\\ &= gcd(s,4054).\end{aligned}$$

Now s and t must be the same parity as they sum to an even number. Moreover we must have $gcd(s,4054)=1$, thus s must be odd. Thus, (s,t) is already in the form (odd, odd) (so it's a hole). Thus all s that satisfies $gcd(s,4054)$ form a solution pair $(s,t)=(s,4054-s)$. So the total count for the number of trajectories

that gets sunk into a hole is given by $\varphi(4054)$ where φ is called Euler's totient function, which counts how many positive integers are co-prime to 4054.

Doing the same for the ball getting sunk in a side pocket, We must have:

$$\begin{aligned} 1 &= gcd(s,t) \\ &= gcd(s, 4053 - s) \\ &= gcd(s, 4053). \end{aligned}$$

But note here, (s,t) could either be $(odd, even)$ or $(even, odd)$ so our final answer for the count of holes of the form $(s,t) = (s, 4053 - s)$ is given by $\frac{\varphi(4053)}{2}$. This is because the pairs that are in the form $(odd, even)$ (which is what we want) form exactly half the total number of (s,t) which satisfy $s + t = 4053$ and are co-prime to each other. Think about why!

So our total number of trajectories which gets sunk in 2026 bounces is given by $4 \cdot (\varphi(4054) + \frac{\varphi(4053)}{2})$. Finally, $4054 = 2 \cdot 2027$, $4053 = 3 \cdot 7 \cdot 193$, so using prime factors to calculate the value of Euler's totient function evaluated at 4054 and 4053, we get:

$$\begin{aligned} \varphi(4054) &= 4054 \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{2027}\right) \\ &= 2026 \\ \varphi(4053) &= 4053 \left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{7}\right)\left(1 - \frac{1}{193}\right) \\ &= 2304 \end{aligned}$$

(Check the Wikipedia page on the totient function for a proof of this above fact).

So our final answer is $4 \cdot (2026 + 1152) = \boxed{12712}$.

Easy right?

So obviously you weren't supposed to do solve this question legit, so how would you arrive at least somewhere close to the right answer?

Well experimenting a bit for how many ways there are to pocket balls in n bounces, we get there are 6 ways for $n = 0$, 12 ways for $n = 1$, 16 ways for $n = 2$, 28 ways for $n = 3$, 28 ways for $n = 4$. We find the growth is roughly linear with a factor of $5 - 7$. If y is our count for number of trajectories where our ball is pocketed after n bounces, we approximate that $y \approx 6n$. Plugging in $n = 2026$, we find $y = 12156$ which is quite close to our original answer. Now obviously there are outliers (notably when $2n + 2$ and $2n + 1$ have very little factors) but approximating the growth as linear will at least put us in the $10,000$ range which is good enough for an estimathon.

4C. How many digits does the 2026th Fibonacci number have? *[The Fibonacci numbers are defined by $F_1 = 1$, $F_2 = 1$, and $F_{n+2} = F_{n+1} + F_n$.]*

SOLUTION

One way to approximate this is via the *Binet formula*

$$F_n = \frac{\varphi^n - (-1)^n \varphi^{-n}}{\sqrt{5}}$$

where $\varphi = \frac{1}{2}(1 + \sqrt{5}) \approx 1.618\ldots$ is the *golden ratio*. We can use the fact that $1 + \lfloor \log(x) \rfloor =$ number of digits of x; we try to approximate $\log F_{2026}$. For large n, the φ^n will dominate, so we can ignore the other term and find

$$\#\text{ of digits} \approx 1 + \log F_{2026} \approx 1 + 2026 \log \varphi - \tfrac{1}{2} \log 5$$

We can ignore $\frac{1}{2} \log 5$ (it's small). Our final answer hinges on a good approximation of $\log \varphi$. Now $\varphi \approx 1.6$, so $\log \varphi \approx \log 16 - \log 10$; $\log 10 = 1$, and $\log 16 = 4 \log 2$. As $\log 2 \approx 0.3$, $\log 16 \approx 1.2$ so $\log \varphi \approx 1.2 - 1 = 0.2$. So our final approximation is

$$\#\text{ of digits} \approx 1 + 2026 \times 0.2 = 406.2$$

so somewhere around 406.2 is a good estimate. In fact, the true answer is $\boxed{424}$.

The 2026th Fibbonacci number is 1,146,764,246,809,443,569,829,753,911,609,133, 976,698,830,606,248,576,472,946,740,393,279,118,624,930,325,957,346,865,657,643, 601,935,723,801,273,921,339,452,870,388,974,473,091,255,329,111,294,089,694,712, 835,772,134,645,299,119,224,720,133,955,543,682,634,678,164,759,302,929,897,481, 725,365,673,661,054,676,508,981,520,225,655,715,452,811,565,366,842,477,711,529, 183,675,753,166,210,543,998,619,738,108,742,529,803,341,435,386,604,375,299,758, 599,667,377,444,018,553,993,205,402,387,551,956,414,766,007,549,577,362,866,254, 371,724,192,911,676,126,716,721,841,143.

INVESTIGATION

So, the number of digits in F_n scales *linearly* with respect to n. How does the number of digits of 2^n scale? Can you come up with a sequence whose number of digits scales quadratically (put differently, the number of digits has a good approximation by a quadratic in n)?